

Final Project Security Devices

Prepared By: ANAS EL MOUHTADI:041178362

OUSSAMA OUAFFAQ:041262019

Site-to-Site IPsec VPN & Firewall Security Configuration

Project Overview

This project involved designing and implementing a secure network infrastructure for a multi-office financial firm (**AC Corp**). The goal was to establish a secure, encrypted communication channel between two remote offices while managing granular internet access security policies using **FortiGate Next-Generation Firewalls (NGFW)**.

Key Objectives

- **Encrypted Tunneling:** Configured a Site-to-Site **IPsec VPN** tunnel between Office 1 (10.0.0.0/24) and Office 2 (172.20.0.0/24) to ensure data confidentiality across the unprotected public internet.
- **Traffic Segmentation & Routing:** Managed dual-ISP connectivity and configured internal routing to ensure local traffic reached the internet via local gateways while inter-office traffic was routed through the secure tunnel.
- **Security Profiles (NGFW):** Implemented advanced security features to meet business requirements:
 - **Antivirus:** Deployed custom AV profiles with deep protocol inspection and "Block" actions for all incoming traffic.
 - **Web/App Filtering:** Restricted social media access (Facebook) in Office 1 and blocked gaming applications (Xbox) in Office 2 to maintain corporate

productivity and security.

- **Connectivity Validation:** Verified end-to-end connectivity using RDP sessions across the tunnel and validated firewall policy enforcement through traffic testing.

Technologies Used

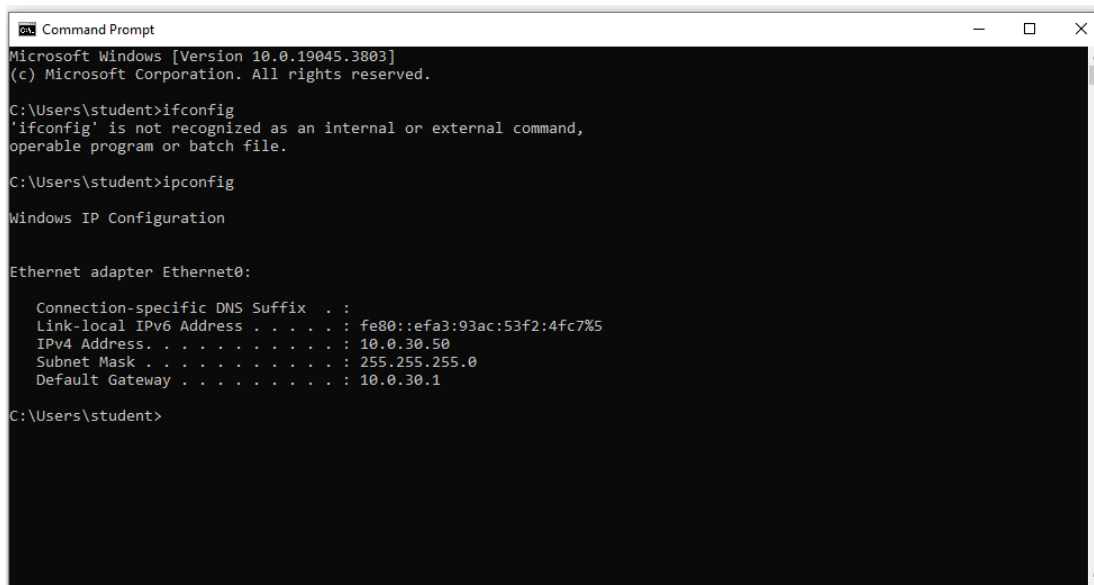
- **Firewall:** FortiGate NGFW (GUI & CLI)
- **VPN Protocol:** IPsec (IKEv2)
- **Networking:** Static Routing, DHCP Server configuration, DNS management
- **Security:** Web Filtering, Application Control, Antivirus Profiles
- **Operating Systems:** Windows (Client), FortiOS

Project deliverables:

Project must be submitted in a PDF format. It should contain name and student ID of team members. Each team member must submit it separately.

Provide screenshots and files below:

- Screenshot of ipconfig command from PC1 and PC2



```
Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\student>ifconfig
'ifconfig' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\student>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::efa3:93ac:53f2:4fc7%5
    IPv4 Address. . . . . : 10.0.30.50
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.30.1

C:\Users\student>
```

```

C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\student>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::b6b4:ac12:8b3:ae9%5
    IPv4 Address. . . . . : 172.20.30.50
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.20.30.1

C:\Users\student>

```

- Screenshot of the Firewall Interfaces from office 1 and 2 (Network > Interfaces)

802.3ad Aggregate						
☐	fortilink	802.3ad Aggregate	Dedicated to FortiSwitch	PING Security Fabric Connection		
Physical Interface 11						
☐	ouaf0001_LAN (port1)	Physical Interface	10.0.30.1/255.255.255.0	PING HTTPS SSH HTTP TELNET		10.0.30.150-10.0.30.200
☐	ouaf0001_Local_ISP (port10)	Physical Interface	192.168.131.129/255.255.255.0	PING		
☐	ouaf0001_WAN_to_Office2 (port9)	Physical Interface	204.5.10.2/255.255.255.0	PING		
☐	port2	Physical Interface	0.0.0.0/0.0.0.0			
☐	port3	Physical Interface	0.0.0.0/0.0.0.0			
☐	port4	Physical Interface	0.0.0.0/0.0.0.0			
☐	port5	Physical Interface	0.0.0.0/0.0.0.0			
☐	port6	Physical Interface	0.0.0.0/0.0.0.0			
☐	port7	Physical Interface	0.0.0.0/0.0.0.0			
☐	port8	Physical Interface	0.0.0.0/0.0.0.0			
Tunnel Interface 1						
☐	NAT interface (nat.root)	Tunnel Interface	0.0.0.0/0.0.0.0			

802.3ad Aggregate						
☐	fortilink	802.3ad Aggregate	Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.255.1.254
Physical Interface 11						
☐	elmo0033_Local_ISP (port10)	Physical Interface	192.168.131.130/255.255.255.0	PING		
☐	elmo0033_WAN_to_Office1 (port9)	Physical Interface	199.100.10.2/255.255.255.0	PING		
☐	ouaf0001_LAN (port1)	Physical Interface	172.20.30.1/255.255.255.0	PING HTTPS SSH HTTP TELNET		172.20.30.150-172.20.30.200
☐	port2	Physical Interface	0.0.0.0/0.0.0.0			
☐	port3	Physical Interface	0.0.0.0/0.0.0.0			
☐	port4	Physical Interface	0.0.0.0/0.0.0.0			
☐	port5	Physical Interface	0.0.0.0/0.0.0.0			
☐	port6	Physical Interface	0.0.0.0/0.0.0.0			
☐	port7	Physical Interface	0.0.0.0/0.0.0.0			
☐	port8	Physical Interface	0.0.0.0/0.0.0.0			
Tunnel Interface 1						
☐	NAT interface (nat.root)	Tunnel Interface	0.0.0.0/0.0.0.0			

- Screenshot of the Firewall Interface setting from office 1 and 2 (Network > Interfaces > Select Interface > Edit)

Edit Interface

Name

ouaf0001_LAN (port1)

Alias

ouaf0001_LAN

Type

Physical Interface

VRF ID ⓘ

0

Role ⓘ

LAN

Address

Addressing mode

Manual IPAM DHCP PPPoE

IP/Netmask

10.0.30.1/255.255.255.0

Create address object matching subnet

☒

Name

port1 address

Destination

10.0.30.0/24

Secondary IP address

☐

Administrative Access

IPv4

☒ HTTPS

☐ FMG-Access

☒ TELNET

☐ Security Fabric

☒ HTTP ⓘ

☒ SSH

☐ FTM

☐ Speed Test

☒ PING

☐ SNMP

☐ RADIUS Accounting

☐ SCIM

Receive LLDP ⓘ

Use VDOM Setting

Enable

Disable

Transmit LLDP ⓘ

Use VDOM Setting

Enable

Disable

☒ DHCP Server

DHCP status

☒ Enabled ☐ Disabled

Address range

10.0.30.150-10.0.30.200

+

Netmask

255.255.255.0

Default gateway

Same as Interface IP

Specify

DNS server

Same as System DNS

Same as Interface IP

Specify

DNS server 1

96.45.45.45

✖

DNS server 2

8.8.8.8

✖

OK

Cancel

Edit Interface

Name  ouaf0001_Local_ISP (port10)
Alias
Type  Physical Interface
VRF ID 
Role 
Estimated bandwidth  kbps Upstream
 kbps Downstream

Address

Addressing mode
Status  Connected
Obtained IP/Netmask 192.168.131.129/255.255.255.0
Expiry Date 2025/12/04 15:21:15
Acquired DNS 192.168.131.2
Default gateway 192.168.131.2
Retrieve default gateway from server 
Distance
Override internal DNS 

Administrative Access

IPv4 ☐ HTTPS ☐ HTTP  ☒ PING
☐ FMG-Access ☐ SSH ☐ SNMP
☐ FTM ☐ RADIUS Accounting ☐ Security Fabric Connection 
☐ Speed Test ☐ SCIM
Receive LLDP 

Traffic Shaping

Outbound shaping profile 
Outbound bandwidth 

Miscellaneous

Comments 0/255

Edit Interface

Name  ouaf0001_WAN_to_Office2 (port9)

Alias

Type  Physical Interface

VRF ID 

Role 

Estimated bandwidth  kbps Upstream

kbps Downstream

☐ Dedicated Management Port

Address

Addressing mode **Manual** IPAM DHCP PPPoE

IP/Netmask

Secondary IP address ☐

Administrative Access

IPv4 ☐ HTTPS ☐ HTTP  ☒ PING

☐ FMG-Access ☐ SSH ☐ SNMP

☐ FTM ☐ RADIUS Accounting ☐ Security Fabric Connection 

☐ Speed Test ☐ SCIM

Receive LLDP  **Enable** **Disable**  View LLDP neighbors

Transmit LLDP  **Enable** **Disable**

Traffic Shaping

Outbound shaping profile ☐

Outbound bandwidth ☐

Miscellaneous

Comments 0/255

Status  **Enabled**  **Disabled**

Edit Interface

Name  elmo0033_Local ISP (port10)

Alias

Type  Physical Interface

VRF ID 

Role 

Estimated bandwidth  kbps Upstream

kbps Downstream

Address

Addressing mode

Status  Connected

Obtained IP/Netmask 192.168.131.130/255.255.255.0

Expiry Date 2025/12/04 15:33:29

Acquired DNS 192.168.131.2

Default gateway 192.168.131.2

Retrieve default gateway from server ☒

Distance

Override internal DNS ☒

Administrative Access

IPv4 ☐ HTTPS ☐ HTTP  ☒ PING

☐ FMG-Access ☐ SSH ☐ SNMP

☐ FTM ☐ RADIUS Accounting ☐ Security Fabric

☐ Speed Test ☐ SCIM ☐ Connection 

Receive LLDP 

Edit Interface

Name	 elmo0033_WAN_to_Office1 (port9)		
Alias	elmo0033_WAN_to_Office1		
Type	 Physical Interface		
VRF ID 	0		
Role 	WAN		
Estimated bandwidth 	0	kbps Upstream	
	0	kbps Downstream	

☐ Dedicated Management Port

Address

Addressing mode	<input checked="" type="button" value="Manual"/> IPAM DHCP PPPoE
IP/Netmask	199.100.10.2/255.255.255.0
Secondary IP address 	

Administrative Access

IPv4	☐ HTTPS	☐ HTTP 	☒ PING
	☐ FMG-Access	☐ SSH	☐ SNMP
	☐ FTM	☐ RADIUS Accounting	☐ Security Fabric Connection 
	☐ Speed Test	☐ SCIM	
Receive LLDP 	Use VDOM Setting <input checked="" type="button" value="Enable"/> Disable View LLDP neighbors		
Transmit LLDP 	Use VDOM Setting <input checked="" type="button" value="Enable"/> Disable		

Traffic Shaping

Outbound shaping profile 
Outbound bandwidth 

Edit Interface

Name  ouaf0001_LAN (port1)

Alias

Type  Physical Interface

VRF ID ⓘ

Role ⓘ

Address

Addressing mode ☒ Manual ☐ IPAM ☐ DHCP ☐ PPPoE

IP/Netmask

Create address object matching subnet ☒

Name  port1 address

Destination

Secondary IP address ☐

Administrative Access

IPv4

☒ HTTPS	☒ HTTP ⓘ	☒ PING
☐ FMG-Access	☒ SSH	☐ SNMP
☒ TELNET	☐ FTM	☐ RADIUS Accounting
☐ Security Fabric Connection ⓘ	☐ Speed Test	☐ SCIM

Receive LLDP ⓘ ☒ Use VDOM Setting ☐ Enable ☐ Disable

Transmit LLDP ⓘ ☒ Use VDOM Setting ☐ Enable ☐ Disable

☒ DHCP Server

DHCP status ☒ Enabled ☐ Disabled

Address range



Netmask

- Screenshot of the VPN tunnel setting from office 1 and 2 (VPN > VPN Tunnels)

Edit VPN Tunnel

Tunnel Settings

Tunnel Objects

Name

elmo0033_toOff1

Comments

Comments

0/255

Network

IP version

IPv4

IPv6

Remote gateway

Static IP address

Dialup user

Dynamic DNS

IP address

204.5.10.2

Interface

elmo0033_WAN_to_Office1 (port...

Local gateway

☐

Mode config

☐

NAT traversal

Enable

Disable

Forced

Dead peer detection

☒

On Idle

On Demand

DPD retry count

3

DPD retry interval

20

second(s)

Forward error correction

☐ Egress

☐ Ingress

EMS SN verification

☒

Advanced network settings

☐

Authentication

Method

Pre-shared Key

Signature

Pre-shared key

Change

IKE

Version 1

Version 2

Mode

Aggressive

Main (ID protection)

XAuth

☐

☐ Phase 1 proposal

Encryption - authentication AES256 - SHA256

+

- Diffie-Hellman groups
- | | |
|-----------------------------|--|
| ☐ 1 | ☐ 2 |
| ☐ 5 | ☒ 14 |
| ☐ 15 | ☐ 16 |
| ☐ 17 | ☐ 18 |
| ☐ 19 | ☐ 20 |
| ☐ 21 | ☐ 27 |
| ☐ 28 | ☐ 29 |
| ☐ 30 | ☐ 31 |
| ☐ 32 | |

Key lifetime 86400 second(s)

Local ID

☐ Phase 2 selectors

+ Create new Edit Delete				
Search				
☐	Name	Local Address	Remote Address	Comments
☐	elmo0033_Phase2_to_Office1	elmo0033_Office2_LAN	elmo033_Office1_LAN	
				1

PN

Edit Phase 2 Selector

rel s

Name

Ne

Comments

0/255

T br

Encapsulation

Tunnel Mode

Transport Mode

ad p

IP version

IPv4

IPv6

PD

Named address

☒

PD

Local address

var

Remote address

SS

Advanced

☐

Adv

set

Include at least one encryption-authentication pair used by the phase 1 proposal.

AES256-SHA256

Encryption - authentication

AES256

-

SHA256

+

Replay detection

Enable

Disable

Perfect forward secrecy (PFS)

Enable

Disable

Diffie-Hellman groups

☐ 1

☐ 2

☐ 5

☐ 14

☐ 15

☐ 16

☐ 17

☐ 18

☐ 19

☒ 20

☒ 21

☐ 27

☐ 28

☐ 29

☐ 30

☐ 31

☐ 32

Local port

All

Specify

Remote port

All

Specify

Protocol

All

Specify

Auto-negotiate

Enable

Disable

Autokey keep alive

Enable

Disable

Key lifetime

Seconds

Kilobytes

Both

43200

second(s)

OK

Cancel

Edit Phase 2 Selector

Name

Comments
0/255

Encapsulation ☒ Tunnel Mode ☐ Transport Mode

IP version ☒ IPv4 ☐ IPv6

Named address ☒

Local address

Remote address

Advanced

Include at least one encryption-authentication pair used by the phase 1 proposal.

- AES256-SHA256

Encryption - authentication -
+

Replay detection ☒ Enable ☐ Disable

Perfect forward secrecy (PFS) ☒ Enable ☐ Disable

Diffie-Hellman groups

☐ 1	☐ 2
☐ 5	☐ 14
☐ 15	☐ 16
☐ 17	☐ 18
☐ 19	☒ 20
☒ 21	☐ 27
☐ 28	☐ 29
☐ 30	☐ 31
☐ 32	

Local port ☒ All ☐ Specify

Remote port ☒ All ☐ Specify

Protocol ☒ All ☐ Specify

Auto-negotiate ☒ Enable ☐ Disable

Autokey keep alive ☒ Enable ☐ Disable

Key lifetime ☒ Seconds ☐ Kilobytes ☐ Both
 second(s)

OK

Cancel

Edit VPN Tunnel

Tunnel Settings Tunnel Objects

Name:

Comments:

0/255

Network

IP version: ☒ IPv4 ☐ IPv6

Remote gateway: ☒ Static IP address ☐ Dialup user ☐ Dynamic DNS

IP address:

Interface:

Local gateway: ☐

Mode config: ☐

NAT traversal: ☒ Enable ☐ Disable ☐ Forced

Keepalive frequency:

Dead peer detection: ☒ On Idle ☒ On Demand

DPD retry count:

DPD retry interval: second(s)

Forward error correction: ☐ Egress ☐ Ingress

EMSSN verification: ☐

Advanced network settings

Authentication

Method: ☒ Pre-shared Key ☐ Signature

Pre-shared key:

IKE: ☒ Version 1 ☐ Version 2

Mode: ☐ Aggressive ☒ Main (ID protection)

XAuth: ☐

Phase 2 selectors

☐	Name	Local Address	Remote Address	Comments
☐	_Phase2_to_Office2	4 elmo0033_Office1_LAN	4 [elmo0033]_Office2_LAN	

Dashboard

Network

Policy & Objects

Security Profiles

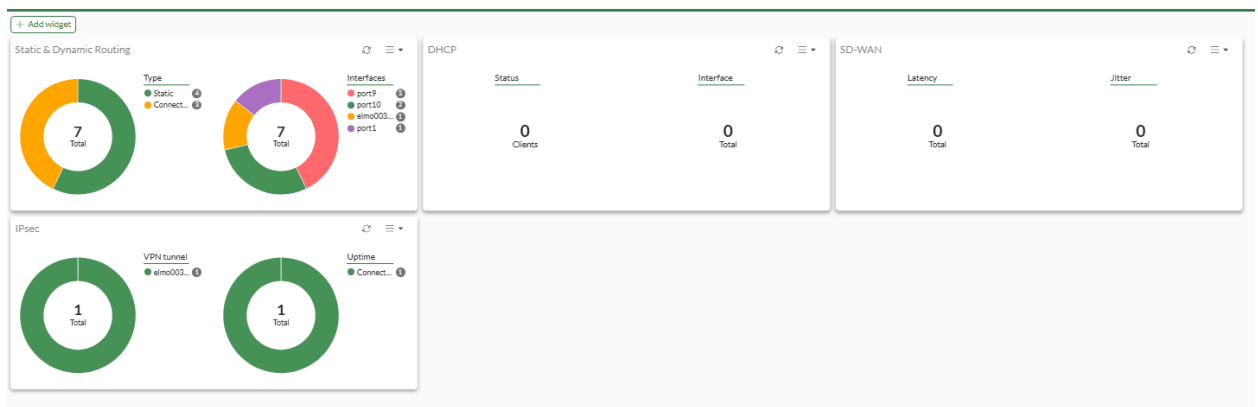
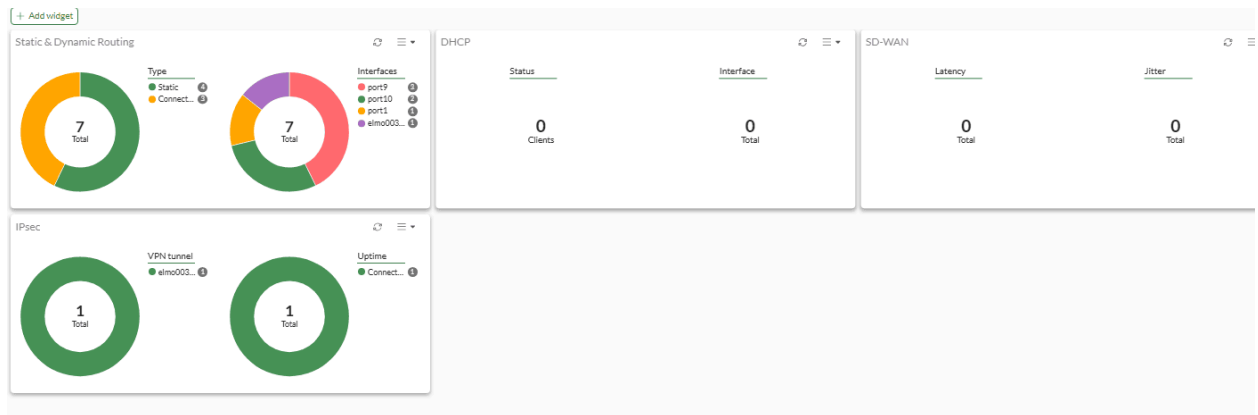
VPN

VPN Wizard

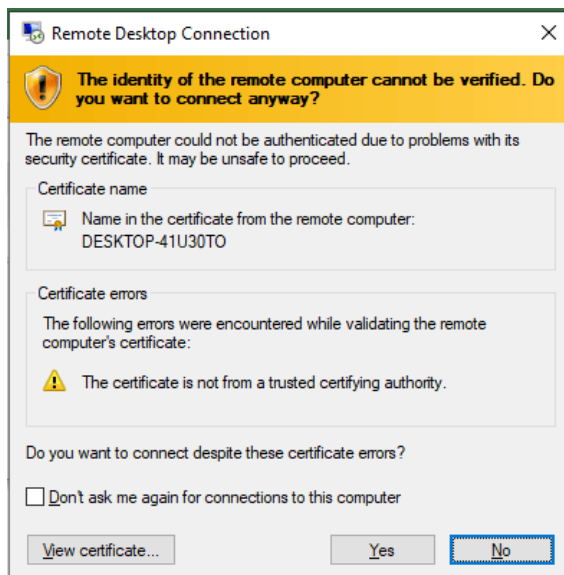
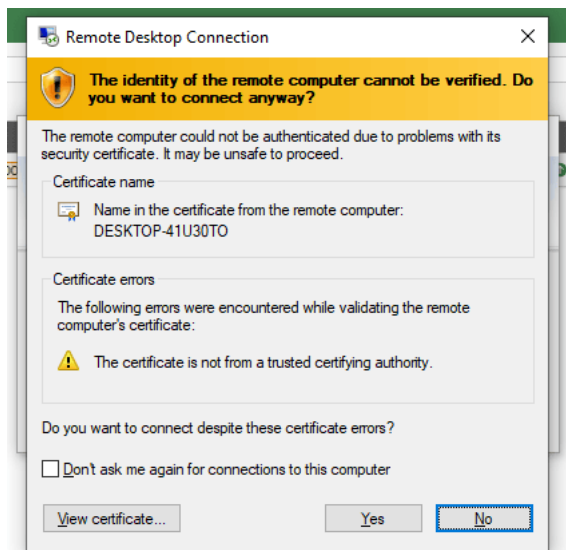
User & Authentication

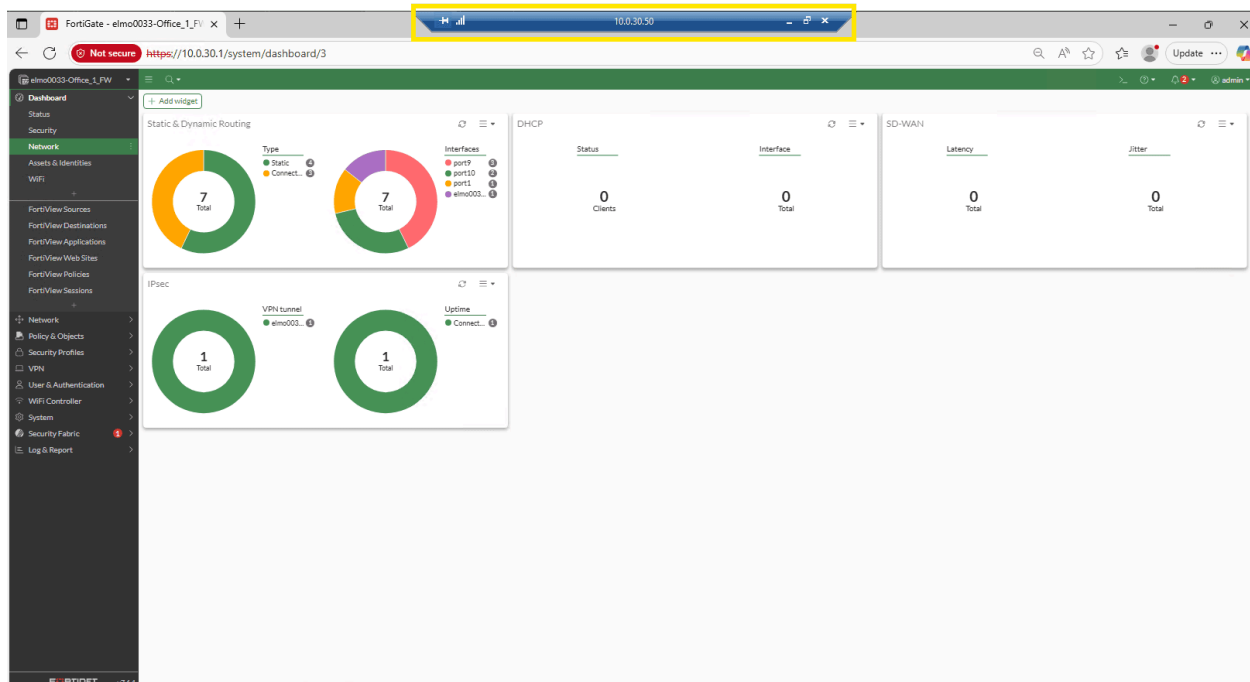
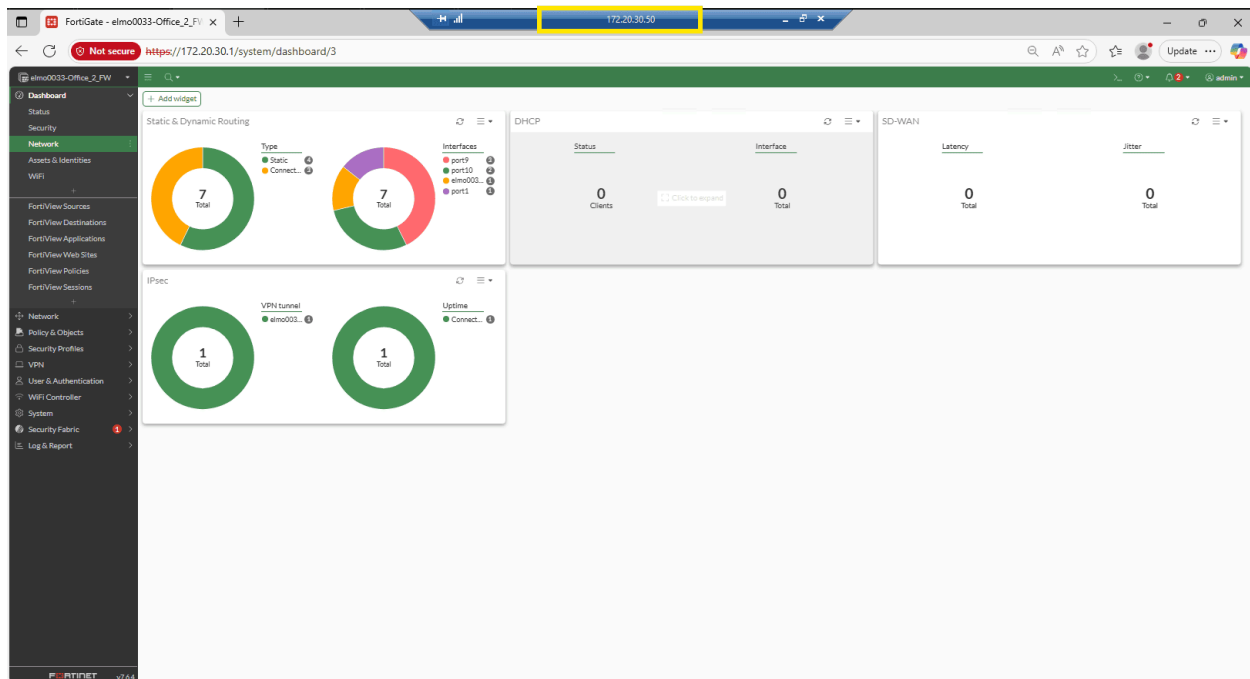
Tunnel	Interface Binding	Status	Ref
☒ Custom	☒ ouaf0001_WAN_to_Office2 (port9)	Up	4

- Screenshot of the IPsec VPN tunnel from office 1 and 2 (Dashboard > Network)



- Screenshot of the Remote Desktop from PC1 to PC2 and vice versa. This screenshot must contain result of ipconfig in both PCs





```
172.20.30.50
Select C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\student>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::b6b4:ac12:8b3:ae9%5
    IPv4 Address. . . . . : 172.20.30.50
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.20.30.1

C:\Users\student>
```

```
10.0.30.50
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\student>ipconfig

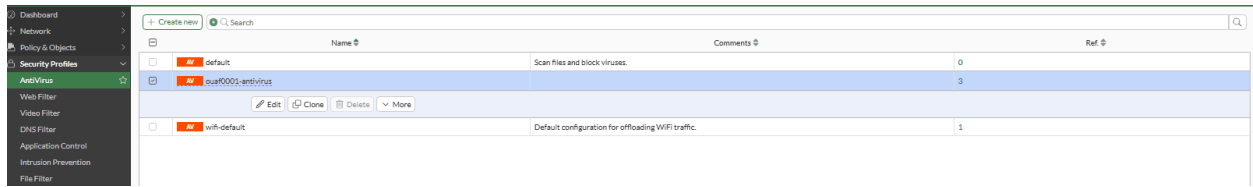
Windows IP Configuration

Ethernet adapter Ethernet0:

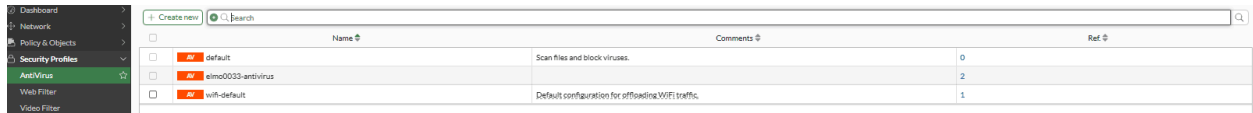
    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::efa3:93ac:53f2:4fc7%5
    IPv4 Address. . . . . : 10.0.30.50
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.30.1

C:\Users\student>
```

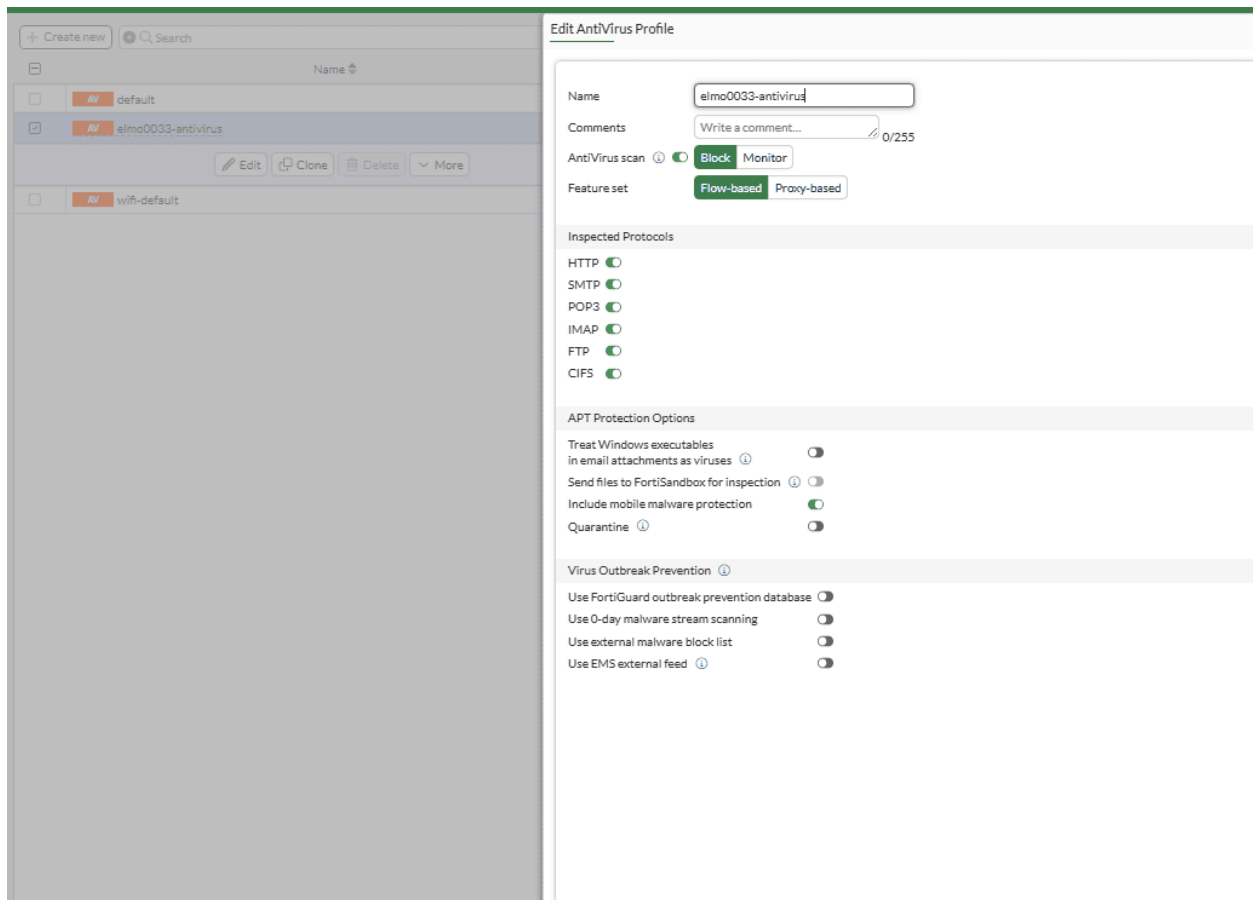
- Screenshot of new Antivirus Profile (Security Profiles > AntiVirus > Select new profile > Edit)



	Name	Comments	Ref
☐	AV default	Scan files and block viruses.	0
☒	AV elmo0001-antivirus		3
Edit Clone Delete More			
☐	AV wi-fi-default	Default configuration for offloading WiFi traffic.	1



	Name	Comments	Ref
☐	AV default	Scan files and block viruses.	0
☒	AV elmo0033-antivirus		2
☐	AV wi-fi-default	Default configuration for offloading WiFi traffic.	1



Edit AntiVirus Profile

Name:

Comments: 0/255

AntiVirus scan: ☒ Block ☐ Monitor

Feature set: ☒ Flow-based ☐ Proxy-based

Inspected Protocols

HTTP ☒
SMTP ☒
POP3 ☒
IMAP ☒
FTP ☒
CIFS ☒

APT Protection Options

Treat Windows executables in email attachments as viruses ☐
Send files to FortiSandbox for inspection ☐
Include mobile malware protection ☒
Quarantine ☐

Virus Outbreak Prevention

Use FortiGuard outbreak prevention database ☒
Use 0-day malware stream scanning ☐
Use external malware block list ☐
Use EMS external feed ☐

+ Create new

🔍 Search

📁

Name 🔍

❑

AV

default

❑

AV

ouaf0001-antivirus

❑

AV

wifi-default

Edit

Clone

Delete

More

Security Rating Insights ⓘ

Edit AntiVirus Profile

Name

ouaf0001-antivirus

Comments

Write a comment...

0/255

AntiVirus scan ⓘ

Block

Monitor

Feature set

Flow-based

Proxy-based

Inspected Protocols

HTTP

SMTP

POP3

IMAP

FTP

CIFS

APT Protection Options

Treat Windows executables in email attachments as viruses ⓘ

Send files to FortiSandbox for inspection ⓘ

Include mobile malware protection

Quarantine ⓘ

Virus Outbreak Prevention ⓘ

Use FortiGuard outbreak prevention database

Use 0-day malware stream scanning

Use external malware block list

Use EMS external feed ⓘ

OK

Cancel

Screenshot of any other security profiles that have been created in each site

Edit Application Sensor

729 Cloud Applications require deep inspection.
2 policies are using this profile. [+](#)

Name

elmo0033-xbox-block

Comments

0/255

Categories

Mixed

All Categories

Business (153 , 11)

Cloud/IT (68 , 2)

Email (75 , 11)

General Interest (238 , 11)

Mobile (3)

Operational Technology

Proxy (189)

Social Media (106 , 27)

Update (48)

VoIP (22)

Unknown Applications

Collaboration (246 , 16)

Game (93)

Generative AI (60 , 611)

Network Service (367)

P2P (52)

Remote Access (98)

Storage/Backup (155 , 24)

Video/Audio (150 , 16)

Web Client (21)

Network Protocol Enforcement

Application and Filter Overrides

+ Create New

Edit

Delete

Priority	Details	Type	Action
1	Xbox Xbox.HTTP	Application	Block

Options

Edit Web Filter Profile

Name

Comments 0/255

Feature set **Flow-based** Proxy-based

FortiGuard Category Based Filter

☒ Allow
 ☒ Monitor
 ☐ Block
 ☐ Warning
 ☐ Authenticate

Name	Action
Local Categories	
custom1	☐ Disable
custom2	☐ Disable
Potentially Liable	
Drug Abuse	☒ Monitor
Hacking	☒ Monitor
Illegal or Unethical	☒ Monitor
Discrimination	☒ Monitor
Explicit Violence	☒ Monitor

0%

☐ Allow users to override blocked categories

Search Engines

Enforce 'Safe Search' on Google, Yahoo!, Bing, Yandex ☒

Static URL Filter

Block invalid URLs ☒

URL Filter ☒

URL	Type	Action	Status
*facebook.com	Wildcard	☐ Block	☒ Enable

Block malicious URLs discovered by FortiSandbox ☐

Content Filter ☐

Rating Options

Behavior when FortiGuard is unreachable ☐ Allow all websites ☒ Block all websites

Rate URLs by domain and IP Address ☐

Proxy Options

HTTP POST Action ☐ Allow ☒ Block

Remove Cookies ☐

Additional Information

[API Preview](#)

[References](#)

[Edit in CLI](#)

Online Guides

[Relevant Docum](#)

[Video Tutorials](#)

Fortinet Communi

[Join the Discussi](#)

Edit Web

Edit URL Filter

Name

Comm

Featur

For

Out

cust

Dr

Hac

Illeg

Disc

Exp

All

Se

Enforc

St

Block

URL F

fa

Block

Conte

Ra

Behavi

Rate U

Pr

HTTP

Remov

URL

*facebook.com

Type

Simple

Regular Expression

Wildcard

Action

☒ Block

Blocks attempts to access any URL that matches the specified URL input.

☐ Allow

The traffic is allowed through the URL filter without being logged. Traffic is subject to further UTM inspection.

☐ Monitor

The traffic is logged, allowed through the URL filter and is subject to further UTM inspection.

☐ Exempt

Traffic is allowed and exempt from further UTM inspection listed below.

Status

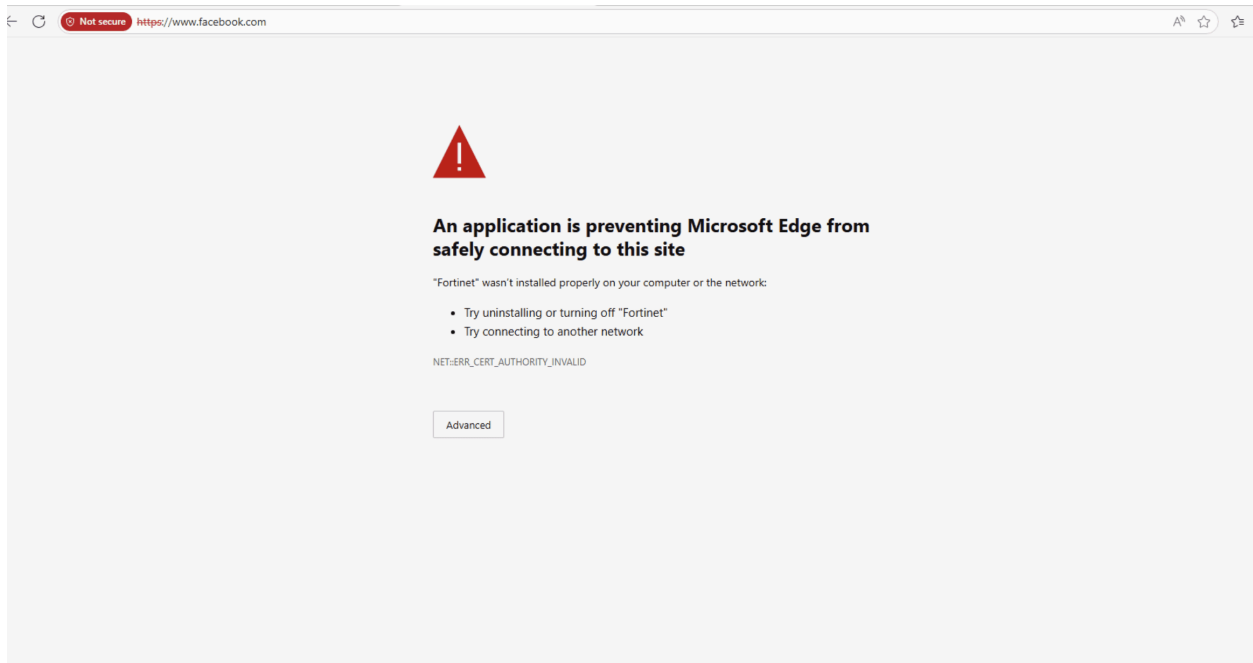
Enable

Disable

OK

Cancel

- Screenshot of unsuccessful access to www.facebook.com from office 1



- Screenshot of unsuccessful access to any “Xbox Game” application from office 2

